Inflation Dynamics - Computational Notebook

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Abstract

This book presents computation details of estimating inflation dynamics.

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So far we have covered estimation of

1. GARCH
2. EGARCH
3. GJR-GARCH
4. SPARCH
5. Regime-switching GARCH
6. BEGE GARCH

State variable definitions:

1. π_t: inflation.
2. $\mathcal{F}_{t-1}$: filtration at time $t-1$.
3. SPF_{t-1}: professional forecast on π_t inflation measured by $\mathcal{F}_{t-1}$.
4. $\hat{\pi}_t$: expected inflation $E[\pi_t \mid \mathcal{F}_{t-1}]$.
5. u_t: inflation innovation (residual), defined as $u_t := \pi_t - \hat{\pi}_t$.
6. u_t^+ : postive residual, defined as $u_t^+ := u_t \cdot \mathbf{1}(u_t > 0)$
7. u_t^- : negative residual, defined as $u_t^- := u_t \cdot \mathbf{1}(u_t \leq 0)$

Computation Notebook

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1. Inflation Summary Statistics

I start with simple statistical summary by plotting inflation and professional forecast over time.

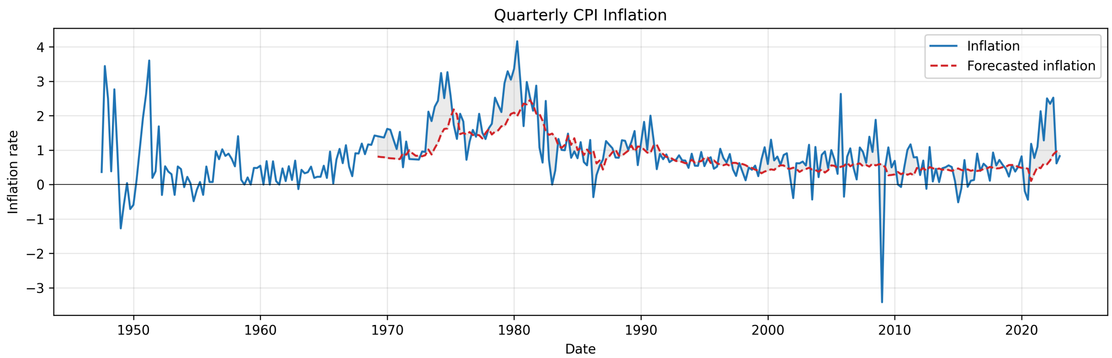


Figure 1: Quarterly inflation π_t and profesional forcast SPF_{t-1}.

Notice that inflation was recorded from 1947 but profesional forecast started from 1969.

	Inflation	Forecasted inflation
Date Start	1947Q2	1969Q1
Date End	2022Q4	2022Q4
#Observations	303	216

Table 1: Quaterly Inflation Statistics

1.a. Effective Sample

To make log-likelihood values comparable across specifications, I use an effective sample for all mean models and all volatility models.

In ARX models, lagged regressors (for example, SPF_{t-1}) are not observed at the very beginning of the raw sample. Therefore, I start the estimation sample at the first date where both current forecast inflation (SPF_t), and one-lag forecast inflation (SPF_{t-1}) are available. For the quarterly data, forecast inflation (SPF_t) is available from **1969Q1** to **2022Q4** (216 observations), requiring one lag (SPF_{t-1}) shifts the usable start to **1969Q2**. So the trimmed estimation window is **1969Q2–2022Q4**, with **215 observations**. This same trimmed sample is then used for all mean specifications, so the residual series has the same length across models.

For GARCH-type recursion, the first few conditional variances are unobserved (for example, σ_0^2, σ_{-1}^2, u_0^2, u_{-1}^2). I initialize these pre-sample variances using the model-implied unconditional variance, for example in GARCH model:

$$\bar{\sigma}^2 = \frac{\omega}{1 - \sum_i \alpha_i - \sum_j \beta_j}, \tag{1}$$

and set

$$u_0^2 = u_{-1}^2 = \cdots = \sigma_0^2 = \sigma_{-1}^2 = \cdots = \bar{\sigma}^2. \tag{2}$$

1.b. Effective Sample Summary Statistics

I report the summary statistics for trimmed effect sample. Skewness and Kurtosis are adjusted by sample size.

	Inflation (π)	SPF	$\pi - SPF$
Date Start	1969Q2	1969Q2	1969Q2
Date End	2022Q4	2022Q4	2022Q4
#Observations	215	215	215
Mean	0.9918	0.8320	0.1598
Median	0.7937	0.6296	0.1101
Std	0.8580	0.4969	0.6642
Min	−3.4170	0.1048	−4.0117
Max	4.1612	2.4566	2.1729
P5	−0.0681	0.3482	−0.7745
P25	0.5328	0.4748	−0.1669
P75	1.2913	1.0150	0.4617
P95	2.5883	2.0030	1.2581
Skewness	0.3550	1.3633	−0.7131
Kurtosis	3.8995	1.2053	7.3006
AC(1)	0.5981	0.9717	0.2764
AC(2)	0.5353	0.9473	0.1322
AC(4)	0.4708	0.8929	0.0881
AC(12)	0.2918	0.6949	0.0011

Table 2: Inflation statistics

2. Mean Process Specification

Four types of mean processes are of interest:

- **Constant**: $\pi_{t+1} = SPF_t + \mu_{t+1}$
- **ARX(1,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,2)**[1]: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \phi_2 SPF_{t-1} + \mu_{t+1}$

where μ_{t+1} is the residual term.

2.a. OLS Results

Estimation results of four mean models with HAC standard errors in parentheses below each estimate. (The log-likelihood is computed under the Gaussian assumption.)

Constant:

$$\hat{\pi}_{t+1} = SPF_t \tag{3}$$

No estimated coefficients. Residuals $\mu_{t+1} = \pi_{t+1} - SPF_t$.

- Log-likelihood: −**222.664**
- AIC: **447.329**, BIC: **450.700**
- Residual skewness: −**0.713**, excess kurtosis: **7.301**

ARX(1,1):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0824} \; \underset{(0.112)}{+0.3005\,\pi_t} \; \underset{(0.167)}{+0.7337\,SPF_t} \tag{4}$$

- Log-likelihood: −**207.381**
- AIC: **420.762**, BIC: **430.874**
- Residual skewness: −**1.047**, excess kurtosis: **8.210**

ARX(2,1):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0897} \; \underset{(0.098)}{+0.2892\,\pi_t} \; \underset{(0.126)}{+0.0834\,\pi_{t-1}} \; \underset{(0.251)}{+0.6385\,SPF_t} \tag{5}$$

- Log-likelihood: −**206.810**
- AIC: **421.619**, BIC: **435.102**
- Residual skewness: −**1.191**, excess kurtosis: **9.050**

ARX(2,2):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0856} \; \underset{(0.106)}{+0.2992\,\pi_t} \; \underset{(0.125)}{+0.0914\,\pi_{t-1}} \; \underset{(0.433)}{+0.4084\,SPF_t} \; \underset{(0.297)}{+0.2136\,SPF_{t-1}} \tag{6}$$

- Log-likelihood: −**206.660**
- AIC: **423.319**, BIC: **440.172**
- Residual skewness: −**1.212**, excess kurtosis: **9.129**

[1] In the GARCH family estimation exercise, we found that ARX(2,2) is always dominated by other three specifications. Thus, we stop estimating ARX(2,2) in BEGE estimation to saving computational resources.

3. GARCH Family

This section presents three GARCH-type volatility models — GARCH, GJR-GARCH, and EGARCH — each combined with one of three conditional distributions for residuals: normal, Student's t, and a finite mixture of normals.

Let $\{u_t\}_{t=1}^{T}$ denote the residual process, and let $\mathcal{F}_{t-1}$ denote the information set available at time $t-1$. We assume

$$u_t = \sigma_t \, z_t, \qquad z_t \mid \mathcal{F}_{t-1} \overset{\text{i. i. d.}}{\sim} D(0,1), \tag{7}$$

where $\sigma_t^2 = \operatorname{Var}(u_t \mid \mathcal{F}_{t-1})$ is the conditional variance and $D(0,1)$ is a standardized distribution (normal, Student's t, or mixture of normals) with zero mean and unit variance. Throughout, $z_t = u_t / \sigma_t$ denotes the standardized residual and θ denotes the full parameter vector (mean process, volatility process parameters together with any distributional parameters).

3.a. *Volatility Recursion Specifications*

GARCH(p, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \, u_{t-i}^2 + \sum_{k=1}^{q} \beta_k \, \sigma_{t-k}^2, \tag{8}$$

with $\omega > 0$, $\alpha_i \geq 0$, $\beta_k \geq 0$, and $\sum_i \alpha_i + \sum_k \beta_k < 1$ for stationarity.

GJR-GARCH(p, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \left(u_{t-i}^{+}\right)^2 + \sum_{j=1}^{p} \gamma_j \left(u_{t-j}^{-}\right)^2 + \sum_{k=1}^{q} \beta_k \, \sigma_{t-k}^2, \tag{9}$$

with $\omega > 0$, $\alpha_i \geq 0$, $\gamma_j \geq 0$, and $\beta_k \geq 0$. The asymmetry coefficient γ_j captures the leverage effect: when $\alpha_j > \gamma_j$, positive shocks raise future volatility more than negative shocks of equal magnitude.

Under the assumption that z_t is symmetric

$$\mathbb{E}\left[\left(u_{t-i}^{+}\right)^2 \mid \mathcal{F}_{t-i-1}\right] = \mathbb{E}\left[\left(u_{t-i}^{-}\right)^2 \mid \mathcal{F}_{t-i-1}\right] = \frac{1}{2}\sigma_{t-i}^2 \tag{10}$$

, the stationarity condition is

$$\frac{1}{2}\sum_{i=1}^{p} \alpha_i + \frac{1}{2}\sum_{j=1}^{p} \gamma_j + \sum_{k=1}^{q} \beta_k < 1. \tag{11}$$

EGARCH(p, q):

$$\ln \sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i (\mid z_{t-i} \mid - \mathbb{E} \mid z_{t-i} \mid) + \sum_{j=1}^{p} \gamma_j \, z_{t-j} + \sum_{k=1}^{q} \beta_k \ln \sigma_{t-k}^2, \tag{12}$$

where $z_t = u_t / \sigma_t$ is the standardized residual. The term α_i captures the magnitude effect (response to the size of shocks), while γ_j captures the sign / leverage effect. Modeling $\ln \sigma_t^2$ guarantees positivity of σ_t^2 without sign restrictions on $\left(\alpha_i, \gamma_j, \beta_k\right)$, and the recursion is covariance-stationary in $\ln \sigma_t^2$ whenever

$$\sum_{k=1}^{q} \beta_k < 1. \tag{13}$$

The constant $\mathbb{E} \mid z_t \mid$ depends on the assumed distribution of the standardized innovation:

- **Standard normal**, $z_t \sim \mathcal{N}(0,1)$:

$$\mathbb{E} \mid z_t \mid = \sqrt{\frac{2}{\pi}}. \tag{14}$$

- **Standardized Student's t** with $\nu > 2$ degrees of freedom (rescaled to unit variance):

$$\mathbb{E} \mid z_t \mid = \frac{2\sqrt{\nu - 2}\,\Gamma\left(\frac{\nu+1}{2}\right)}{(\nu - 1)\sqrt{\pi}\,\Gamma\left(\frac{\nu}{2}\right)}. \tag{15}$$

As $\nu \to \infty$ this collapses to the normal value $\sqrt{2/\pi}$; for finite ν it is strictly larger, reflecting the heavier tails.

- **Mixture of two normals** with parameters (p_1, μ_1, σ_1^2) and induced (p_2, μ_2, σ_2^2) (as defined in the likelihood section): using $\mathbb{E} \mid X \mid = \mu\,[2\Phi(\mu/s) - 1] + 2s\,\phi(\mu/s)$ for $X \sim \mathcal{N}(\mu, s^2)$ (a folded-normal mean), where Φ and ϕ are the standard normal CDF and PDF,

$$\mathbb{E} \mid z_t \mid = \sum_{m=1}^{2} p_m[\,\mu_m(2\Phi(\mu_m/\sigma_m) - 1) + 2\sigma_m\,\phi(\mu_m/\sigma_m)]. \tag{16}$$

EGARCH(p, q):

$$\ln\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i\left(\mid z_{t-i} \mid - \sqrt{\frac{2}{\pi}}\right) + \sum_{j=1}^{p} \gamma_j\,z_{t-j} + \sum_{k=1}^{q} \beta_k \ln\sigma_{t-k}^2, \tag{17}$$

where $z_t = u_t/\sigma_t$ is the standardized residual. The term α_i captures the magnitude effect (response to the size of shocks) while γ_j captures the sign / leverage effect. Modeling $\ln\sigma_t^2$ guarantees positivity of σ_t^2 without sign restrictions on $(\alpha_i, \gamma_j, \beta_k)$, and the recursion is covariance-stationary in $\ln\sigma_t^2$ whenever

$$\sum_{k=1}^{q} \beta_k < 1. \tag{18}$$

Remark on the centering constant. The constant $\sqrt{2/\pi}$ equals $\mathbb{E} \mid z_t \mid$ when $z_t \sim \mathcal{N}(0,1)$, so under a Gaussian innovation the term $\mid z_{t-i} \mid - \sqrt{2/\pi}$ has zero conditional mean and ω admits the clean interpretation

$$\mathbb{E}[\ln\sigma_t^2] = \frac{\omega}{1 - \sum_{k=1}^{q} \beta_k}. \tag{19}$$

Following the convention of the `arch` Python package, we retain $\sqrt{2/\pi}$ as a fixed constant in the recursion even when the innovation distribution is Student's t or a mixture of normals. Under those distributions, $\mathbb{E} \mid z_t \mid \neq \sqrt{2/\pi}$, so the centered term $\mid z_{t-i} \mid - \sqrt{2/\pi}$ no longer has zero mean and ω loses the unconditional-mean interpretation above. This choice is observationally innocuous: replacing $\sqrt{2/\pi}$ with the distribution-specific constant $c_D = \mathbb{E}_D \mid z_t \mid$ yields an equivalent model whose fitted $\hat{\alpha}_i, \hat{\gamma}_j, \hat{\beta}_k$ are unchanged, with only the intercept shifted by

$$\omega = \omega^* + \left(\sqrt{\frac{2}{\pi}} - c_D\right) \sum_{i=1}^{p} \alpha_i. \tag{20}$$

The likelihood, the fitted σ_t^2 path, and all forecasts are identical under the two parameterizations.

3.b. *Log-Likelihood Specifications*

For each candidate distribution we give (i) the standardized density of z_t, (ii) the conditional density of $u_t \mid \mathcal{F}_{t-1}$ obtained via the change of variables $u_t = \sigma_t z_t$ (which contributes a Jacobian factor $1/\sigma_t$), (iii) the per-observation log-likelihood $\ell_t(\theta)$, and (iv) the sample log-likelihood

$$\ln L(\theta) = \sum_{t=1}^{T} \ell_t(\theta). \tag{21}$$

3.b.i. *Normal distribution:*

The standardized innovation satisfies $z_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0,1)$ with density

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\}. \tag{22}$$

By the change of variables $u_t = \sigma_t z_t$, the conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1}; \theta) = \frac{1}{\sigma_t} \phi\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sqrt{2\pi\, \sigma_t^2}} \exp\left\{-\frac{u_t^2}{2\, \sigma_t^2}\right\}. \tag{23}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\frac{1}{2}\left(\ln 2\pi + \ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2}\right), \tag{24}$$

and the sample log-likelihood is

$$\ln L(\theta) = -\frac{1}{2} \sum_{t=1}^{T} \left(\ln 2\pi + \ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2}\right). \tag{25}$$

3.b.ii. *Student's t distribution:*

Let $\nu > 2$ denote the degrees-of-freedom parameter. The standardized Student's t density (rescaled to unit variance) is

$$f_\nu(z) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)}} \left(1 + \frac{z^2}{\nu-2}\right)^{-\frac{\nu+1}{2}}, \tag{26}$$

where $\Gamma(\cdot)$ is the gamma function. The restriction $\nu > 2$ ensures finite variance. The conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1}; \theta) = \frac{1}{\sigma_t} f_\nu\left(\frac{u_t}{\sigma_t}\right) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)\, \sigma_t^2}} \left(1 + \frac{u_t^2}{(\nu-2)\, \sigma_t^2}\right)^{-\frac{\nu+1}{2}}. \tag{27}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = \ln\Gamma\left(\frac{\nu+1}{2}\right) - \ln\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\ln\left(\ \pi(\nu-2)\sigma_t^2\ \right) - \frac{\nu+1}{2}\ln\left(1 + \frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right), \tag{28}$$

and the sample log-likelihood is

$$\ln L(\theta) = T\left[\ln\Gamma\left(\frac{\nu+1}{2}\right) - \ln\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\ln\left(\ \pi(\nu-2)\ \right)\right] - \frac{1}{2}\sum_{t=1}^{T}\ln\sigma_t^2 - \frac{\nu+1}{2}\sum_{t=1}^{T}\ln\left(1 + \frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right). \tag{29}$$

3.b.iii. *Mixture of two normals:*

The standardized innovation follows a two-component Gaussian mixture with parameters (p_1, μ_1, σ_1^2):

$$f_{\text{mix}}(z) = p_1\,\phi(z;\mu_1,\sigma_1^2) + p_2\,\phi(z;\mu_2,\sigma_2^2), \qquad \phi(z;\mu,s^2) = \frac{1}{\sqrt{2\pi s^2}}\exp\left\{-\frac{(z-\mu)^2}{2s^2}\right\}. \tag{30}$$

To enforce $\mathbb{E}[z_t] = 0$ and $\text{Var}(z_t) = 1$, the remaining parameters are determined by

$$p_2 = 1 - p_1, \qquad \mu_2 = -\frac{p_1\mu_1}{p_2}, \qquad \sigma_2^2 = \frac{1 - p_1(\sigma_1^2 + \mu_1^2) - p_2\mu_2^2}{p_2}. \tag{31}$$

This leaves three free parameters: (p_1, μ_1, σ_1^2).

Hierarchical representation. The mixture admits the latent-variable form $S \sim \text{Bernoulli}(p_1)$, $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$, with $z = X_1$ if $S = 1$ and $z = X_2$ otherwise. As a consequence, moments and the CDF decompose as

$$\mathbb{E}[z^n] = p_1\,\mathbb{E}[X_1^n] + p_2\,\mathbb{E}[X_2^n], \qquad F_{\text{mix}}(a) = p_1\,\Phi\left(\frac{a-\mu_1}{\sigma_1}\right) + p_2\,\Phi\left(\frac{a-\mu_2}{\sigma_2}\right), \tag{32}$$

where Φ is the standard normal CDF.

The conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1};\theta) = \frac{1}{\sigma_t}f_{\text{mix}}\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sigma_t}\left[p_1\,\phi(z_t;\mu_1,\sigma_1^2) + p_2\,\phi(z_t;\mu_2,\sigma_2^2)\right], \qquad z_t = \frac{u_t}{\sigma_t}. \tag{33}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\ln\sigma_t + \ln\left[p_1\,\phi(z_t;\mu_1,\sigma_1^2) + p_2\,\phi(z_t;\mu_2,\sigma_2^2)\right], \tag{34}$$

and the sample log-likelihood is

$$\ln L(\theta) = -\sum_{t=1}^{T}\ln\sigma_t + \sum_{t=1}^{T}\ln\left[p_1\,\phi(z_t;\mu_1,\sigma_1^2) + p_2\,\phi(z_t;\mu_2,\sigma_2^2)\right]. \tag{35}$$

3.b.iv. *Gaussian Mixture distribution:*

The mixture of 2 normal distribution given parameters p_1,μ_1,σ_1^2 is

$$f(z_t) = p_1\frac{1}{\sqrt{2\pi\sigma_1^2}}exp\left\{-\frac{\left(z_t-\mu_1\right)^2}{2\sigma_1^2}\right\} + p_2\frac{1}{\sqrt{2\pi\sigma_2^2}}exp\left\{-\frac{\left(z_t-\mu_2\right)^2}{2\sigma_2^2}\right\} \tag{36}$$

where $p_2 = 1 - p_1$,$\mu_2 = \frac{-p_1\mu_1}{p_2}$, and $\sigma_2^2 = \frac{1-p_2\mu_2^2-p_1\left(\sigma_1^2+\mu_1^2\right)}{p_2}$

N th order moment can be calculated as:

$$E[z^n] = \int z^n \left[p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} exp\left\{ -\frac{(z-\mu_1)^2}{2\sigma_1^2} \right\} + (1-p_1) \frac{1}{\sqrt{2\pi\sigma_2^2}} exp\left\{ -\frac{(z-\mu_2)^2}{2\sigma_2^2} \right\} \right] dz \quad (37)$$

$$E[z^n] = p_1 E[x_1^n \mid x_1 \sim N(\mu_1, \sigma_1^2)] + p_2 E[x_2^n \mid x_2 \sim N(\mu_2, \sigma_2^2)] \quad (38)$$

Similary, CDF can be calculated analytically by

$$\Phi(a) = \int_{-\inf}^{a} z \left[p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} exp\left\{ -\frac{(z-\mu_1)^2}{2\sigma_1^2} \right\} + p_2 \frac{1}{\sqrt{2\pi\sigma_2^2}} exp\left\{ -\frac{(z-\mu_2)^2}{2\sigma_2^2} \right\} \right] dz \quad (39)$$

$$\Phi(a) = p_1 \Phi_{x_1}(a) + p_2 \Phi_{x_2}(a) \quad (40)$$

We can think mixture of two normal distributions generated by three random variable Z,X_1,X_2. Z follows Bernoulli(P_1), $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$. In order to generate a random sample point, we can first use Z to decide which normal distributions to use, and then randomly pick a point in that normal distribution.

Log likelihood function is

$$L(u_t) = \sum_{t=1}^{T} \ln \frac{1}{\sigma_t} \left[p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} exp\left\{ -\frac{(z_t-\mu_1)^2}{2\sigma_1^2} \right\} + p_2 \frac{1}{\sqrt{2\pi\sigma_2^2}} exp\left\{ -\frac{(z_t-\mu_2)^2}{2\sigma_2^2} \right\} \right] \quad (41)$$

, where $z_t = \frac{u_t}{\sigma_t}$

3.c. Estimation Summary

3.d. Best Estimation Illustrations

Table 2: Model selection by AIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	(1,1)	(2,1)	359.4710	(1,1)	(1,1)	368.2429
t	(1,1)	(2,1)	343.1431	(2,1)	(1,1)	344.6254
Skew *t*	(1,1)	(2,1)	345.1312	(2,1)	(1,1)	346.5791
GED	(1,1)	(2,1)	345.1925	(1,1)	(1,1)	347.6934
Mix of Normal	(1,1)	(2,1)	345.8222	(2,1)	(2,1)	343.5023

Table 3: Model selection by BIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	FC	(2,1)	381.4280	FC	(1,1)	383.4754
t	FC	(1,1)	367.8350	FC	(1,1)	367.1732
Skew *t*	FC	(1,1)	370.4242	FC	(1,1)	369.6649
GED	FC	(1,1)	371.8489	FC	(1,1)	370.9644
Mix of Normal	FC	(1,1)	373.7106	FC	(1,1)	371.0689

4. Model Selection Report

Table 2: Model selection by AIC: GARCH vs GJR vs EGARCH

Distribution	GARCH Mean	GARCH Vol	GARCH AIC	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	(1,1)	(2,1)	387.6789	(1,1)	(2,1)	372.8425	(1,1)	(2,1)	374.2995
t	(2,1)	(2,1)	362.6718	(1,1)	(2,1)	357.3153	(1,1)	(2,1)	356.7451
Mix of Normal	(2,1)	(1,2)	365.6060	(2,1)	(2,1)	359.0690	(1,1)	(2,1)	356.4744

Table 3: Model selection by BIC: GARCH vs GJR vs EGARCH

Distribution	GARCH Mean	GARCH Vol	GARCH BIC	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	FC	(2,1)	401.4156	FC	(2,1)	396.6581	FC	(2,1)	398.1066
t	FC	(2,1)	387.7770	FC	(1,1)	384.2113	FC	(1,1)	383.8115
Mix of Normal	(1,1)	(2,1)	401.6779	FC	(2,1)	391.9670	FC	(1,1)	385.2567

4.a. Best Models by Criterion and Volatility Family

For each criterion and each volatility family, the selected model is the best across mean-process choices, orders, and distributions using stable & successful fits.

4.a.i. *AIC:*

GARCH:

- Model ID: `M079`
- Distribution: `$t$`
- Mean process: `(2,1)` (ARX_2_1)
- Volatility process: `GARCH(2,1)`

$$\hat{\pi}_{t+1} = 0.0754 + 0.2410\,\pi_t + 0.1539\,\pi_{t-1} + 0.5650\,SPF_t + \mu_{t+1} \tag{42}$$

Robust SE: `Const` (0.0804), `Inflation_lag_1` (0.1565), `Inflation_lag_2` (0.2555), `SPF` (0.4729).

$$\hat{\sigma}_t^2 = 0.1369 + 0.4194\,u_{t-1}^2 + 0.3468\,u_{t-2}^2 + 0.0588\,\sigma_{t-1}^2 \tag{43}$$

- Number of observations: **215**
- Log-likelihood: **−172.335915**
- AIC: **362.671831**, BIC: **393.007573**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
Const	0.075394	0.080383	0.937930	0.348280
Inflation_lag_1	0.241000	0.156521	1.539733	0.123625

Parameter	Coef	Std Err	t-value	p-value
Inflation_lag_2	0.153880	0.255527	0.602208	0.547036
SPF	0.565016	0.472887	1.194822	0.232156
alpha[1]	0.419351	0.260560	1.609418	0.107525
alpha[2]	0.346759	1.053861	0.329037	0.742128
beta[1]	0.058760	2.090575	0.028107	0.977577
nu	4.472871	1.614077	2.771164	0.005586
omega	0.136928	0.470167	0.291233	0.770873

GJR-GARCH:

- Model ID: `M068`
- Distribution: `$t$`
- Mean process: `(1,1)` (ARX_1_1)
- Volatility process: `GJR-GARCH(2,1)`

$$\hat{\pi}_{t+1} = 0.0846 + 0.1984\,\pi_t + 0.7847\,SPF_t + \mu_{t+1} \tag{44}$$

Robust SE: `Const` (0.0816), `Inflation_lag_1` (0.0996), `SPF` (0.1317).

$$\hat{\sigma}_t^2 = 0.1168 + 0.4252\,(u_{t-1}^+)^2 + 0.2174\,(u_{t-1}^-)^2 + 0.6575\,(u_{t-2}^+)^2 + 0.0000\,(u_{t-2}^-)^2 + 0.1354\,\sigma_{t-1}^2 \tag{45}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

- Number of observations: **215**
- Log-likelihood: **−168.657674**
- AIC: **357.315348**, BIC: **391.021728**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
Const	0.084561	0.081554	1.036868	0.299797
Inflation_lag_1	0.198379	0.099598	1.991802	0.046393
SPF	0.784679	0.131672	5.959354	0.000000
alpha[1]	0.425191	0.219441	1.937612	0.052671
alpha[2]	0.657467	0.374661	1.754833	0.079288
beta[1]	0.135397	0.172961	0.782816	0.433735
gamma[1]	−0.207829	0.251448	−0.826528	0.408504
gamma[2]	−0.657467	0.305776	−2.150159	0.031543
nu	5.349660	2.200946	2.430618	0.015073
omega	0.116818	0.042038	2.778885	0.005455

EGARCH:

- Model ID: `M117`
- Distribution: `Mix of Normal`

- Mean process: `(1,1)` (ARX_1_1)
- Volatility process: `EGARCH(2,1)`

$$\hat{\pi}_{t+1} = 0.0861 + 0.2148\,\pi_t + 0.7325\,SPF_t + \mu_{t+1} \tag{46}$$

Robust SE: `Const` (0.0317), `Inflation_lag_1` (0.0705), `SPF` (0.0017).

$$\ln \hat{\sigma}_t^2 = -0.3146 + 0.5764\left(\mid z_{t-1} \mid - \sqrt{2/\pi}\right) + 0.0053\left(\mid z_{t-2} \mid - \sqrt{2/\pi}\right) + 0.1006\,z_{t-1} + 0.3061\,z_{t-2} + 0.6993\ln \sigma_{t-1}^2 \tag{47}$$

- Number of observations: **215**
- Log-likelihood: **−166.237213**
- AIC: **356.474426**, BIC: **396.922083**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
Const	0.086145	0.031694	2.718014	0.006568
Inflation_lag_1	0.214824	0.070517	3.046416	0.002316
SPF	0.732514	0.001739	421.149922	0.000000
alpha[1]	0.576377	0.147631	3.904164	0.000095
alpha[2]	0.005343	0.238386	0.022413	0.982118
beta[1]	0.699319	0.105518	6.627490	0.000000
gamma[1]	0.100565	0.099839	1.007278	0.313801
gamma[2]	0.306148	0.099347	3.081593	0.002059
mu_1	0.048865	0.057085	0.855992	0.392002
omega	−0.314557	0.158733	−1.981672	0.047516
p_1	0.934951	0.041199	22.693639	0.000000
sigma_1_sq	0.625460	0.155173	4.030724	0.000056

4.a.ii. *BIC:*

GARCH:

- Model ID: `M055`
- Distribution: `$t$`
- Mean process: `FC` (Constant_anchor)
- Volatility process: `GARCH(2,1)`

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{48}$$

No mean-process coefficients are estimated in this anchored specification.

$$\hat{\sigma}_t^2 = 0.1221 + 0.4616\,u_{t-1}^2 + 0.4956\,u_{t-2}^2 + 0.0000\,\sigma_{t-1}^2 \tag{49}$$

- Number of observations: **215**
- Log-likelihood: **−180.461891**
- AIC: **370.923781**, BIC: **387.776971**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
alpha[1]	0.461596	0.179330	2.574004	0.010053
alpha[2]	0.495589	0.229219	2.162077	0.030612
beta[1]	0.000000	0.151356	0.000000	1.000000
nu	4.807722	1.636402	2.937984	0.003304
omega	0.122062	0.045238	2.698212	0.006971

GJR-GARCH:

- Model ID: `M050`
- Distribution: `$t$`
- Mean process: `FC` (Constant_anchor)
- Volatility process: `GJR-GARCH(1,1)`

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{50}$$

No mean-process coefficients are estimated in this anchored specification.

$$\hat{\sigma}_t^2 = 0.0636 + 0.7929\,\left(u_{t-1}^{+}\right)^2 + 0.1465\,\left(u_{t-1}^{-}\right)^2 + 0.4366\,\sigma_{t-1}^2 \tag{51}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

- Number of observations: **215**
- Log-likelihood: **−178.679065**
- AIC: **367.358130**, BIC: **384.211320**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
alpha[1]	0.792917	0.256260	3.094187	0.001974
beta[1]	0.436615	0.116958	3.733109	0.000189
gamma[1]	−0.646461	0.231689	−2.790208	0.005267
nu	4.760505	1.533912	3.103507	0.001912
omega	0.063611	0.025467	2.497746	0.012499

EGARCH:

- Model ID: `M051`
- Distribution: `$t$`
- Mean process: `FC` (Constant_anchor)
- Volatility process: `EGARCH(1,1)`

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{52}$$

No mean-process coefficients are estimated in this anchored specification.

$$\ln\hat{\sigma}_t^2 = -0.2520 + 0.5595\left(\mid z_{t-1}\mid - \sqrt{2/\pi}\right) + 0.2663\,z_{t-1} + 0.7882\ln\sigma_{t-1}^2 \tag{53}$$

- Number of observations: **215**
- Log-likelihood: **−178.479161**

- AIC: **366.958322**, BIC: **383.811512**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
alpha[1]	0.559468	0.139250	4.017715	0.000059
beta[1]	0.788161	0.061181	12.882450	0.000000
gamma[1]	0.266302	0.077913	3.417950	0.000631
nu	5.151108	1.764823	2.918768	0.003514
omega	−0.251989	0.105679	−2.384477	0.017103

4.b. EGARCH and Initialization Notes

- Effective sample is fixed at 215 observations for all models (`hold_back=0` with explicit lag regressors in the mean equation).
- In `arch`, EGARCH uses the package's centered term `|z|-sqrt(2/pi)` in the recursion for all distributions.
- Under non-Gaussian errors (e.g., Student's `t` or mixture), this is an intercept reparameterization; fitted dynamics and likelihood are still valid.
- The recursion start (`backcast`) is updated iteratively during estimation in this script: fit `->` implied initial variance from current parameters `->` refit.
- Stability is monitored using a persistence proxy (`<1`): GARCH uses `sum(alpha)+sum(beta)`, GJR uses `sum(alpha)+0.5*sum(gamma)+sum(beta)`, EGARCH uses `sum(beta)`.

5. Regime-Switching Inflation Models

Let the latent state be

$$s_t \in \{1, ..., K\}, \qquad \Pr(s_t = j \mid s_{t-1} = i) = p_{ij}. \tag{54}$$

I estimate several specifications that differ in **which blocks of parameters are allowed to switch with s_t**. Taking the ARX(2,2) model as an example,

$$\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 \, \mathrm{SPF}_t + \phi_2 \, \mathrm{SPF}_{t-1} + \mu_{t+1}, \tag{55}$$

the parameters are grouped into three blocks, and each block can independently be made regime-dependent:

Block	State-dependent parameters
AR component	$c_{s_t},\ \rho_{1,s_t},\ \rho_{2,s_t}$
SPF component	$\phi_{1,s_t},\ \phi_{2,s_t}$
Shock distribution	$\sigma_{s_t},\ \nu_{s_t}$

where ν_{s_t} denotes the degrees of freedom when assuming standardized Student-t distribution for residuals. A given specification is defined by selecting any subset of these three blocks to be switching; parameters in blocks not selected are held constant across regimes.

Mean	Dist	K	Sw.AR	Sw.SPF	Sw.Var	Sw.nu	LogLik	AIC	BIC
Constant	Normal	2	N	N	Y	N	−186.63	381.26	394.743
Constant	Normal	3	N	N	Y	N	−180.655	379.311	409.646
Constant	Student t	2	N	N	Y	Y	−185.7	383.4	403.624
Constant	Student t	3	N	N	Y	Y	−180.389	384.778	425.226
ARX(1,1)	Normal	2	N	N	Y	N	−177.189	368.378	391.972
ARX(1,1)	Normal	2	Y	N	Y	N	−175.118	368.235	398.571
ARX(1,1)	Normal	2	Y	Y	Y	N	−175.092	370.185	403.891
ARX(1,1)	Normal	3	N	N	Y	N	−173.724	371.448	411.896
ARX(1,1)	Normal	3	Y	N	Y	N	−162.374	356.747	410.677
ARX(1,1)	Student t	2	N	N	Y	Y	−175.494	368.988	399.324
ARX(1,1)	Student t	2	Y	N	Y	Y	−162.972	347.945	385.022
ARX(1,1)	Student t	2	Y	Y	Y	Y	−162.961	349.921	390.369
ARX(1,1)	Student t	3	N	N	Y	Y	−173.091	376.182	426.742
ARX(1,1)	Student t	3	Y	N	Y	Y	−155.151	348.301	412.343
ARX(2,1)	Normal	2	N	N	Y	N	−176.216	368.431	395.396
ARX(2,1)	Normal	2	Y	N	Y	N	−174.942	371.884	408.961
ARX(2,1)	Normal	2	Y	Y	Y	N	−174.377	372.754	413.201
ARX(2,1)	Normal	3	N	N	Y	N	−171.151	368.301	412.119
ARX(2,1)	Normal	3	Y	N	Y	N	−156.72	351.44	415.482
ARX(2,1)	Student t	2	N	N	Y	Y	−173.425	366.849	400.556
ARX(2,1)	Student t	2	Y	N	Y	Y	−160.571	347.142	390.96
ARX(2,1)	Student t	2	Y	Y	Y	Y	−159.534	347.068	394.257
ARX(2,1)	Student t	3	N	N	Y	Y	−171.069	374.139	428.069
ARX(2,1)	Student t	3	Y	N	Y	Y	−150.126	344.253	418.407
ARX(2,2)	Normal	2	N	N	Y	N	−175.961	369.922	400.258
ARX(2,2)	Normal	2	Y	N	Y	N	−174.534	373.067	413.515
ARX(2,2)	Normal	2	Y	Y	Y	N	−174.155	376.31	423.499
ARX(2,2)	Normal	3	N	N	Y	N	−171.029	370.057	417.246
ARX(2,2)	Normal	3	Y	N	Y	N	−156.663	353.325	420.738
ARX(2,2)	Student t	2	N	N	Y	Y	−173.233	368.466	405.543
ARX(2,2)	Student t	2	Y	N	Y	Y	−160.567	349.135	396.324
ARX(2,2)	Student t	2	Y	Y	Y	Y	−157.588	347.176	401.107
ARX(2,2)	Student t	3	N	N	Y	Y	−170.754	375.507	432.808
ARX(2,2)	Student t	3	Y	N	Y	Y	−149.466	344.932	422.457

6. Best Regime-Switching Model

Selected by AIC among converged models.

6.a. Model Summary

- Model label: `ARX(2,1)_student_t_r3_arY_spfN`
- Mean process: `ARX(2,1)`
- Mean equation target: `Inflation`
- Intercept included: `Y`
- Intercept switches by regime: `Y`
- Error distribution: `Student t`
- #Regime: `3`
- Switching AR: `Y`
- Switching SPF: `N`
- Switching distribution variance: `Y`
- Switching nu: `Y`
- Sample: `1969Q2 to 2022Q4`
- Nobs: `215`
- LogLik: `-150.126`
- AIC: `344.253`
- BIC: `418.407`

6.b. Mean Model Specification

```
Inflation = c_(s_t) + beta_{Inflation_lag_1}(s_t) * Inflation_lag_1 +
beta_{Inflation_lag_2}(s_t) * Inflation_lag_2 + beta_{SPF} * SPF + u_t
```

Regime-specific mean coefficients:

regime	Intercept	Inflation_lag_1	Inflation_lag_2	SPF
0	0.872769	−0.54292	−0.238057	0.574951
1	0.381796	0.0492982	−0.0989449	0.574951
2	−0.0670625	0.26872	0.448281	0.574951

6.c. Mean Process Parameters

parameter	estimate
const[0]	0.872769
const[1]	0.381796
const[2]	−0.0670625
Inflation_lag_1[0]	−0.54292
Inflation_lag_1[1]	0.0492982
Inflation_lag_1[2]	0.26872
Inflation_lag_2[0]	−0.238057
Inflation_lag_2[1]	−0.0989449
Inflation_lag_2[2]	0.448281

parameter	estimate
SPF	0.574951

6.d. Distribution Parameters

parameter	estimate
sigma2[0]	0.0118058
sigma2[1]	0.612973
sigma2[2]	0.208789
nu[0]	183.071
nu[1]	2.39744
nu[2]	104.326

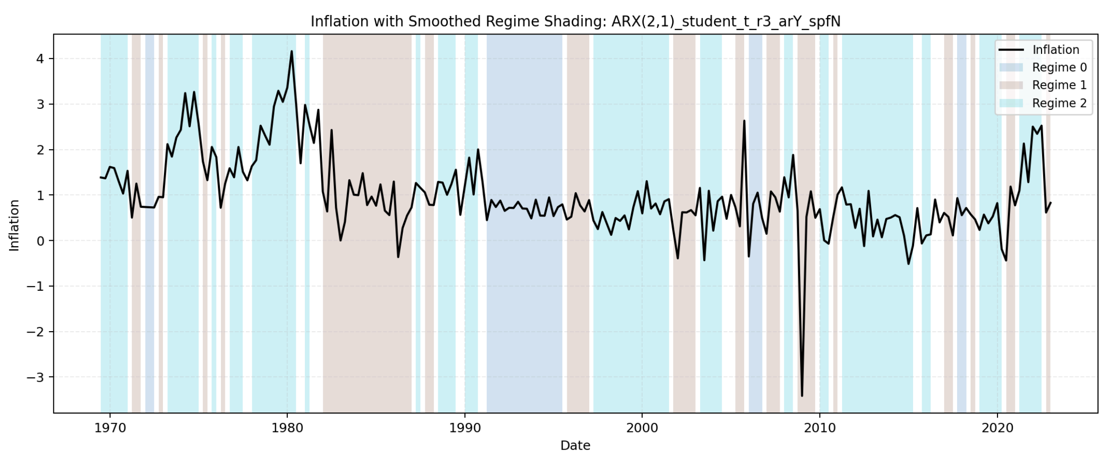
6.e. Transition Probabilities (Vector Form)

parameter	estimate
p[0->0]	0.832798
p[1->0]	0.0480447
p[2->0]	0.000150046
p[0->1]	0.165512
p[1->1]	0.646779
p[2->1]	0.24607

6.f. Transition Probability Matrix

From \ To	Regime 0	Regime 1	Regime 2	Row Sum
Regime 0	0.832798	0.165512	0.00168973	1
Regime 1	0.0480447	0.646779	0.305177	1
Regime 2	0.000150046	0.24607	0.75378	1

6.g. Regime Classification Plot



7. BEGE Density

BEGE density function $f_{BEGE}(\mu \mid p, n, \sigma_p, \sigma_n)$ is the function that calculates the density of the observation μ given the parameters $\{p, n, \sigma_p, \sigma_n\}$.

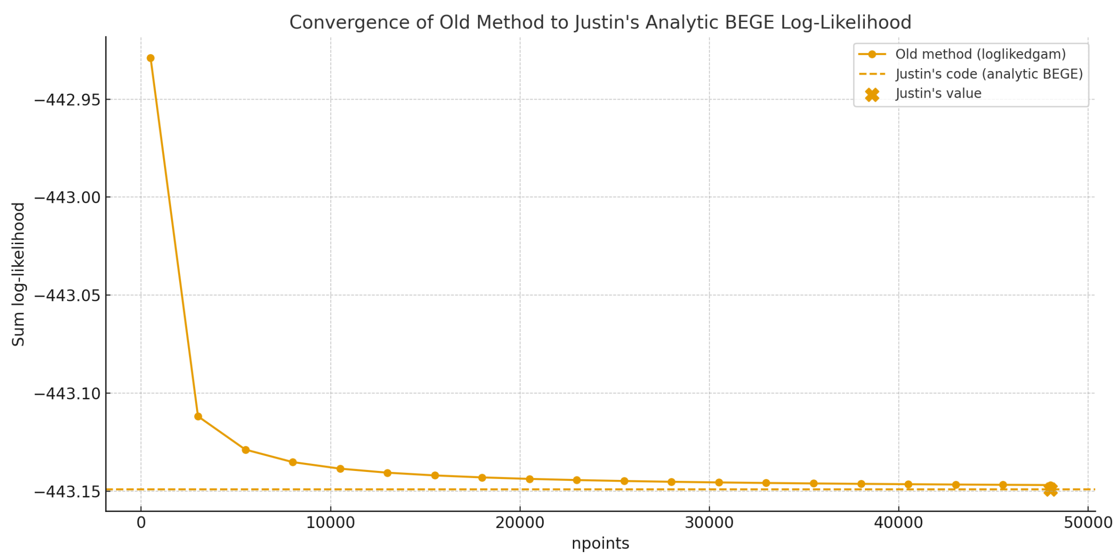
To compare Justin's analytic BEGE density code with the old numerical one, I conducted an experiment using synthetic data. I generated 200 independent observations from a standard normal distribution and fixed the BEGE parameters to

$$p = 5, \qquad n = 7, \qquad \sigma_p = 0.8, \qquad \sigma_n = 1.2. \tag{56}$$

I computed the sum of log-likelihood using two approaches:

1. The *old BEGE function*, which approximates the density via numerical integration on a uniform grid, and whose accuracy depends strongly on the number of grid points; and
2. *Justin's analytic BEGE function*.

The old numerical method was evaluated over a wide range of grid resolutions (up to $50,000$ points), and the resulting log-likelihood sums were plotted alongside the benchmark value computed from Justin's code.



Findings:

- The old numerical likelihood converges toward Justin's analytic likelihood as the number of grid points increases, but the convergence is slow and requires very fine grids.
- For small or moderate grid sizes, the numerical integration *overestimates* the log-likelihood. This explains why, for a given set of parameters, Justin's implementation produces a lower log-likelihood than the old code.
- Justin's analytic BEGE formula provides a more accurate and more computationally efficient evaluation.

8. BEGE GARCH Family

To start the random search of initial mean parameters, I draw uniform samples from $(\mu - 2\sigma,\ \mu + 2\sigma)$ where μ and σ are mean and standard deviation from OLS regression. I also set the AR coefficient bound to avoid the AR process to explode. We have four types of mean processes.

Table 1: Parameter Bounds for Mean Process Specifications

Model	c	ρ_1	ρ_2	ϕ_1	ϕ_2
Constant	–	–	–	–	–
ARX(1,1)	$(\min \pi_t,\ \ \max \pi_t)$	$(-0.999,\ 0.999)$	–	$(-10,\ 10)$	–
ARX(2,1)	$(\min \pi_t,\ \ \max \pi_t)$	$(-1.999,\ 1.999)$	$(-0.999,\ 0.999)$	$(-10,\ 10)$	–
ARX(2,2)	$(\min \pi_t,\ \ \max \pi_t)$	$(-1.999,\ 1.999)$	$(-0.999,\ 0.999)$	$(-10,\ 10)$	$(-10,\ 10)$

8.a. Estimation Speed Controls

The BEGE likelihood now uses the analytic hypergeometric-U density for moderate shape paths and a cumulant-generating-function saddlepoint density for large recursive shape states. This avoids the numerical overflow/cancellation that can make direct `hyperu` evaluation report impossible log densities when p_t or n_t is large. Exact high-precision `hyperu` checks can still be forced with `hyperu_method="mpmath"` or the estimator-specific `density_hyperu_method="mpmath"`.

For the multi-start BEGE estimators, robust numerical standard errors are optional through `compute_se`. The default is `compute_se=False` for fast model search; the result collectors recompute standard errors for the reported best AIC rows after selection.

8.b. Best-Model Reporting Screen

Raw BEGE search outputs are kept in `output/raw/draw_###.csv`. The collectors recompute the likelihood from each stored parameter vector using the stabilized BEGE density before writing `results/all_estimations.csv`, so stale likelihoods from earlier density code do not determine the best model. Reported best-model tables require finite corrected likelihood criteria, successful optimizer convergence, finite positive shape paths, positive conditional variance paths, and the documented parameter, stability, and unconditional-variance constraints.

The diagnostic value $\max_t \{p_t, n_t\}$ is recorded but is not used as a selection exclusion rule. The companion `results/selection_diagnostics.csv` records stored versus corrected likelihood criteria, high-shape density usage, implied persistence, minimum scale, maximum shape path, and the exclusion reason for each saved estimation row. The reported best-AIC rows with standard errors are written to `results/best_aic_with_se.csv`.

9. BEGE Density

This audit compares three fixed-shape BEGE density implementations on the ARX(1,1) residuals from the canonical effective sample.

The ARX(1,1) residual is computed as

$$u_t = \pi_t - (c + \rho_1 \pi_{t-1} + \phi_1 SPF_t), \tag{57}$$

using the OLS coefficients re-estimated on the 1969Q2–2022Q4 sample:

$$\hat{\pi}_t = 0.082431 + 0.300540 \pi_{t-1} + 0.733720 SPF_t. \tag{58}$$

The Gaussian OLS log likelihood is `-207.381`, AIC is `420.762`, and BIC is `430.874`.

9.a. ARX(1,1) Residual Summary

Statistic	Value
Date start	1969Q2
Date end	2022Q4
Observations	215
Mean	0.000000
Std	0.636326
Min	−4.131190
P5	−0.945215
Median	−0.015764
P95	1.006587
Max	2.078000
Skewness	−1.047441
Excess kurtosis	8.210174

9.b. Shape Parameter Range From Previous BEGE Runs

The range below uses eligible ARX(1,1) rows from the BadGood BEGE results only. For each saved parameter vector I recomputed the recursive shape paths on the common ARX(1,1) residuals, then summarized all observations and all eligible rows. The density comparison itself keeps the selected shape values constant across time.

Model	Rows	p q05	p median	p q95	n q05	n median	n q95	sigma_p med	sigma_n med
BadGood	994	1.4285	2.8130	6.4330	0.0714	0.3320	2.8342	0.2355	0.5555

9.c. Fixed-Shape Parameter Sets

The log likelihood and timing comparison uses the following five BadGood-based parameter sets. In each row, `sigma_p` and `sigma_n` are held at their BadGood medians.

Set	p	n	sigma_p	sigma_n
p q05, n median	1.428538	0.331970	0.235491	0.555532

Set	p	n	sigma_p	sigma_n
p median, n median	2.813004	0.331970	0.235491	0.555532
p q95, n median	6.433015	0.331970	0.235491	0.555532
p median, n q05	2.813004	0.071355	0.235491	0.555532
p median, n q95	2.813004	2.834194	0.235491	0.555532

9.d. Log Likelihood Comparison

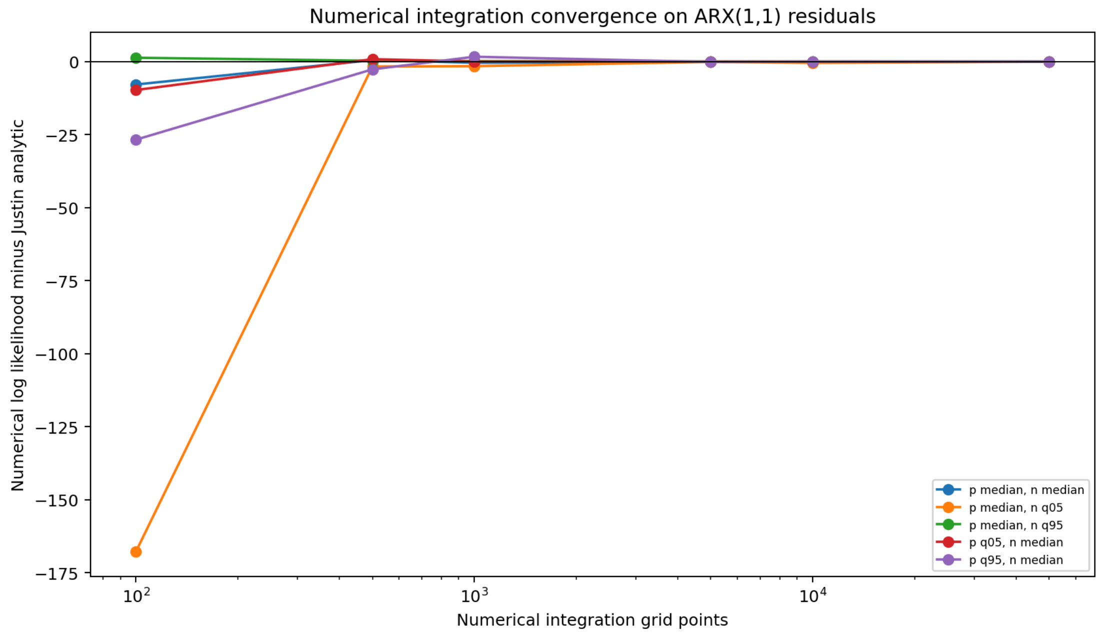
`BEGE_density.py` is the current implementation. `BEGE_density_Justin.py` is Justin's analytic formula, and the numerical columns use `BEGE_density_Numerical_Integration.py::loglikedgam_constant` at the stated grid size.

Parameter set	My function	Justin function	Numerical 100	Numerical 500	Numerical 1000	Numerical 5000	Numerical 10000	Numerical 50000
p q05, n median	−214.162839	−214.162839	−223.900355	−213.330306	−214.116656	−214.148282	−214.146613	−214.160080
p median, n median	−188.957117	−188.957117	−196.801762	−188.438308	−189.359597	−189.033173	−188.947127	−188.954400
p q95, n median	−194.340481	−194.340481	−221.076698	−196.996227	−192.679189	−194.464527	−194.368904	−194.347495
p median, n q05	−212.877473	−212.877473	−380.654962	−214.556763	−214.448495	−212.976377	−213.323667	−212.904257
p median, n q95	−235.887717	−235.887717	−234.563988	−235.609633	−235.748009	−235.859932	−235.873999	−235.885265

9.e. Evaluation Speed Comparison

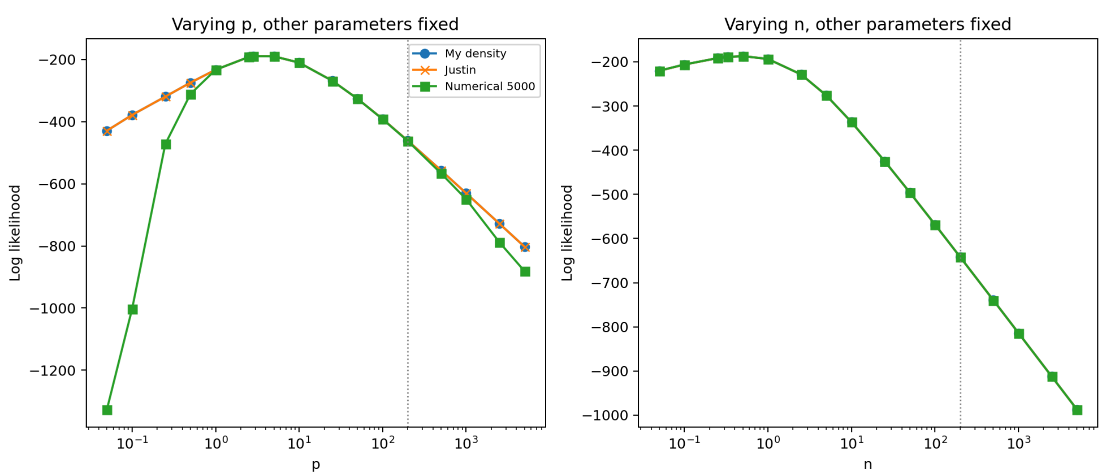
Parameter set	My function	Justin function	Numerical 100	Numerical 500	Numerical 1000	Numerical 5000	Numerical 10000	Numerical 50000
p q05, n median	0.0003	0.0005	0.0086	0.0427	0.0883	0.4326	0.9102	4.5478
p median, n median	0.0004	0.0006	0.0075	0.0358	0.0730	0.3674	0.7613	3.8117
p q95, n median	0.0003	0.0005	0.0064	0.0308	0.0663	0.3258	0.6520	3.2415
p median, n q05	0.0003	0.0005	0.0076	0.0358	0.0737	0.3567	0.7479	3.6527

Parameter set	My function	Justin function	Numerical 100	Numerical 500	Numerical 1000	Numerical 5000	Numerical 10000	Numerical 50000
p median, n q95	0.0004	0.0006	0.0059	0.0284	0.0611	0.2900	0.5937	2.9724



9.f. *Shape Tail Consistency*

Holding the other parameters at the BadGood medians (`p=2.8130`, `n=0.3320`, `sigma_p=0.2355`, `sigma_n=0.5555`), I vary either `p` or `n` up to 5000 and compare the three density functions. The numerical integration line uses 5,000 grid points.



9.g. *Findings*

- BadGood quantiles provide the fixed-shape examples used in the likelihood and timing tables.

- The numerical integration function is useful as a convergence diagnostic, but it is much slower than the analytic functions and remains grid-sensitive.
- The shape-tail plot varies one shape parameter at a time while holding the other shape and both scale parameters at the BadGood medians.

Generated audit files:

- `Density_Analysis/BEGE_density_shape_ranges.csv`
- `Density_Analysis/BEGE_density_representative_sets.csv`
- `Density_Analysis/BEGE_density_comparison.csv`
- `Density_Analysis/BEGE_density_numerical_grid.csv`
- `Density_Analysis/BEGE_density_llf_table.csv`
- `Density_Analysis/BEGE_density_speed_table.csv`
- `Density_Analysis/BEGE_density_consistency.csv`

10. Constant BEGE

BEGE with invariant shape parameters is

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1} \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n \end{aligned} \tag{59}$$

where

$$\omega_p \sim \tilde{\Gamma}(\bar{p}, 1), \quad \omega_n \sim \tilde{\Gamma}(\bar{n}, 1). \tag{60}$$

$\bar{p}$ and $\bar{n}$ are unconditional shape parameters. The random search and bounds are set to be:

- $0.05 < \sigma_p, \sigma_n < 2$
- $0.1 < p, n < 10$

11. Constant BEGE Best Model Summary

Generated: `2026-05-28T19:48:01` Total estimations: `20000` Successful estimations: `19986`

> **Warning**
>
> Excluded 1 successful estimate(s) from best-model selection because `shape_p + shape_n` is numerically an integer, a known unstable point for the SciPy hyperu BEGE-density evaluation. Top excluded row: `constant`, seed `30`, draw `32`, reported LogLik `355.392806`.

11.a. Best by Mean Type (Log-Likelihood)

Mean Type	Seed	Draw	LogLik	AIC	BIC
constant	1	41	−199.843879	407.687759	421.170311
ARX(1,1)	50	48	−184.396177	382.792353	406.386819
ARX(2,1)	33	65	−181.884824	379.769649	406.734753
ARX(2,2)	23	92	−181.688414	381.376828	411.712571

11.b. Parameter Estimates From Best Log-Likelihood Fits

11.b.i. *constant:*

Parameter	Estimate	Std. Error
shape_p	2.675920	0.208030
shape_n	0.185967	0.109333
sigma_p	0.298260	0.086151
sigma_n	1.572583	0.062739

11.b.ii. *ARX(1,1):*

Parameter	Estimate	Std. Error
c	0.056014	0.004172
rho_1	0.323706	0.007503
phi_1	0.737787	0.014056
shape_p	2.627875	0.008489
shape_n	0.281123	0.010973
sigma_p	0.285666	0.006623
sigma_n	0.800204	0.003558

11.b.iii. *ARX(2,1):*

Parameter	Estimate	Std. Error
c	0.039254	0.014593
rho_1	0.316821	0.076697
rho_2	0.189143	0.046505

Parameter	Estimate	Std. Error
phi_1	0.539235	0.027847
shape_p	3.317147	0.037952
shape_n	0.228113	0.064524
sigma_p	0.246532	0.053103
sigma_n	0.928310	0.026420

11.b.iv. *ARX(2,2):*

Parameter	Estimate	Std. Error
c	0.004200	0.003184
rho_1	0.323229	0.002223
rho_2	0.163364	0.002537
phi_1	0.410644	0.002267
phi_2	0.194111	0.001594
shape_p	2.665345	0.002129
shape_n	0.282140	0.002203
sigma_p	0.271156	0.003024
sigma_n	0.840201	0.004885

12. Symmetric BEGE

The model assumes different scaled parameters and different constants in GJR recursions for both components:

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p\,\omega_p - \sigma_n\,\omega_n, \end{aligned} \tag{61}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1). \tag{62}$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $(p_0, n_0, \rho, \phi^+, \phi^-)$:

$$\begin{aligned} p_t &= p_0 + \rho\,p_{t-1} + \frac{\phi^+}{2\sigma_p^2}\left(u_{t-1}^+\right)^2 + \frac{\phi^-}{2\sigma_p^2}\left(u_{t-1}^-\right)^2, \\ n_t &= n_0 + \rho\,n_{t-1} + \frac{\phi^+}{2\sigma_n^2}\left(u_{t-1}^+\right)^2 + \frac{\phi^-}{2\sigma_n^2}\left(u_{t-1}^-\right)^2 \end{aligned} \tag{63}$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$,
- $10^{-5} < \rho < 0.999$,
- $10^{-5} < \phi^+, \phi^- < 0.999$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

The current estimation code enforces the stability condition

$$\rho + \frac{\phi^+}{2} + \frac{\phi^-}{2} < 1, \tag{64}$$

and the same variance guard used by the current BEGE searches,

$$\sigma_p^2 p_0 + \sigma_n^2 n_0 < 0.87. \tag{65}$$

The hard cap $\max\{p_t, n_t\} < 200$ is not imposed during raw multi-start optimization or reported best-model selection by default. The stabilized BEGE density is evaluated directly as long as the recursive shape series are finite; large shape states use the saddlepoint density backend. The collector writes the maximum shape path as a diagnostic in `results/selection_diagnostics.csv`.

13. Batch Estimation

`BEGE_symmetric1.py` estimates the Symmetric BEGE specification on the canonical effective sample from `DataSummary/README.md`, **1969Q2–2022Q4** with **215 observations**. The runner uses the precomputed lag columns when they are available, so lagged ARX regressors do not trim the sample again.

The script estimates the four mean-process specifications:

- Constant
- ARX(1,1)

- ARX(2,1)
- ARX(2,2)

Results are checkpointed to `output/raw/draw_###.csv` after each draw. This keeps completed mean-process rows when a seed job is interrupted, while the final write still contains all requested rows when the seed finishes.

`collect_symmetric_results.py` merges the raw seed files and writes:

- `results/all_estimations.csv`, without standard-error columns or optimizer messages.
- `results/selection_diagnostics.csv`, with stored and corrected likelihood criteria plus selection diagnostics.
- `results/best_aic_with_se.csv`, with standard errors for the corrected best AIC fit in each eligible mean process.
- `results/best_model.md`, with the corrected best AIC/BIC models and reported parameter standard errors.

When `START_ID` and `END_ID` are set, the collector only merges `draw_###.csv` files in that seed range. This prevents older seed files from entering a new smaller resubmission.

`SimulationSymmetric.sh` defaults to 400 seed jobs, 10 draws per mean process, 10 optimizer starts per draw, and 300 SLSQP iterations per start. This gives 40,000 optimizer starts for each mean process. With all four mean processes, that is 160,000 optimizer starts total.

14. Symmetric BEGE Best Model Summary

Generated: `2026-05-31T16:32:31` Total estimations: `4004` Converged estimations: `4004` Eligible estimations for best-model selection: `4004`

Saved likelihoods are recomputed from the stored parameter paths before ranking. Large recursive shape states are evaluated by the BEGE saddlepoint density backend; `max(p_t, n_t)` is reported as a diagnostic, not as an exclusion rule.

Selection screen: finite corrected AIC/BIC/log-likelihood, successful optimizer status, finite positive shape paths, positive conditional variance paths, and documented parameter/stability/unconditional-variance constraints.

14.a. *Global Best by AIC*

- Mean type: `ARX(2,1)`
- Seed / draw: `33` / `3`
- AIC: `336.821887`
- BIC: `373.898906`
- LogLik: `-157.410944`
- Max shape: `321.028674`

14.b. *Global Best by BIC*

- Mean type: `constant`
- Seed / draw: `80` / `4`
- AIC: `348.225798`
- BIC: `371.820264`
- LogLik: `-167.112899`
- Max shape: `509.938159`

14.c. *Eligible Best by Mean Type (AIC)*

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	80	4	348.225798	371.820264	−167.112899	509.938159	0.046945
ARX(1,1)	75	7	360.128702	393.835082	−170.064351	39.977562	0.184266
ARX(2,1)	33	3	336.821887	373.898906	−157.410944	321.028674	0.055808
ARX(2,2)	59	1	361.653630	402.101286	−168.826815	35.324447	0.177234

14.d. *Eligible Best by Mean Type (BIC)*

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	80	4	348.225798	371.820264	−167.112899	509.938159	0.046945
ARX(1,1)	75	7	360.128702	393.835082	−170.064351	39.977562	0.184266
ARX(2,1)	33	3	336.821887	373.898906	−157.410944	321.028674	0.055808
ARX(2,2)	59	1	361.653630	402.101286	−168.826815	35.324447	0.177234

14.e. *Parameter Estimates From Best AIC Fits*

14.e.i. *constant:*

SE status: `computed`

Parameter	Estimate	Std. Error
p0	6.009042	1944.188830
n0	6.566534	1049.350091
rho	0.632466	0.141303
phi_plus	0.301054	16.107963
phi_minus	0.118481	46.852710
sigma_p	0.046945	0.000392
sigma_n	0.063736	0.000816

14.e.ii. *ARX(1,1):*

SE status: `computed`

Parameter	Estimate	Std. Error
c	0.127718	0.084455
rho_1	0.226108	0.081926
phi_1	0.713765	0.117150
p0	1.685365	0.492642
n0	0.050587	0.049623
rho	0.518141	0.159624
phi_plus	0.557483	0.193902
phi_minus	0.044598	0.079540
sigma_p	0.184266	0.049650
sigma_n	0.646891	0.375694

14.e.iii. *ARX(2,1):*

SE status: `computed`

Parameter	Estimate	Std. Error
c	−0.146472	0.055409
rho_1	0.538570	0.012069
rho_2	0.203855	0.005827
phi_1	−0.192917	0.009124
p0	1.761364	0.004706
n0	6.492780	0.482157
rho	0.908778	0.000005
phi_plus	0.051370	0.000009
phi_minus	0.063027	0.002754
sigma_p	0.663130	0.007542

Parameter	Estimate	Std. Error
sigma_n	0.055808	0.000005

14.e.iv. *ARX(2,2):*

SE status: `computed`

Parameter	Estimate	Std. Error
c	0.118775	0.099447
rho_1	0.222018	0.085452
rho_2	0.119321	0.086342
phi_1	0.458211	0.314800
phi_2	0.136363	0.282144
p0	1.489645	1.153155
n0	0.040169	0.039032
rho	0.583471	0.268069
phi_plus	0.465367	0.303378
phi_minus	0.049596	0.090857
sigma_p	0.177234	0.056448
sigma_n	0.730911	0.395504

15. Bad and Good Symmetric GARCH Model

Here n_t, p_t is generated recursively with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p, \phi_n)$:

$$\begin{aligned} p_t &= p_0 + \rho_p \, p_{t-1} + \frac{\phi_p}{2\sigma_p^2} \left(u_{t-1}\right)^2, \\ n_t &= n_0 + \rho_n \, n_{t-1} + \frac{\phi_n}{2\sigma_n^2} \left(u_{t-1}\right)^2 \end{aligned} \tag{66}$$

I have set the constraints below.

- $\rho + \phi < 1$ for both good and bad shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \operatorname{Var}(\pi_t)$.

The hard cap $\max\{p_t, n_t\} < 200$ is not imposed during raw multi-start optimization or reported best-model selection by default. The stabilized BEGE density is evaluated directly as long as the recursive shape series are finite; large shape states use the saddlepoint density backend. The collector writes the maximum shape path as a diagnostic in `results/selection_diagnostics.csv`.

16. Estimation Workflow

`BG_GJR1.py` estimates the BadGood BEGE specification on the canonical effective sample from `DataSummary/README.md`, **1969Q2–2022Q4** with **215 observations**. The runner uses the precomputed lag columns when they are available, so lagged ARX regressors do not trim the sample again.

The script estimates the four mean-process specifications:

- Constant
- ARX(1,1)
- ARX(2,1)
- ARX(2,2)

Results are checkpointed to `output/raw/draw_###.csv` after each draw. This keeps completed mean-process rows when a seed job is interrupted, while the final write still contains all requested rows when the seed finishes.

`collect_bg_results.py` merges the raw seed files and writes:

- `results/all_estimations.csv`, without standard-error columns or optimizer messages.
- `results/selection_diagnostics.csv`, with stored and corrected likelihood criteria plus selection diagnostics.
- `results/best_aic_with_se.csv`, with standard errors for the corrected best AIC fit in each eligible mean process.
- `results/best_model.md`, with the corrected best AIC/BIC models and reported parameter standard errors.

When `START_ID` and `END_ID` are set, the collector only merges `draw_###.csv` files in that seed range. This prevents older seed files from entering a new smaller resubmission.

`SimulationBG.sh` defaults to 400 seed jobs, 10 draws per mean process, 10 optimizer starts per draw, and 300 SLSQP iterations per start. This gives 40,000 optimizer starts for each mean process. With all four mean processes, that is 160,000 optimizer starts total. The `estimate` action prints a rough per-seed time estimate before submitting jobs.

17. BadGood BEGE Best Model Summary

Generated: `2026-05-31T16:30:31` Total estimations: `4020` Converged estimations: `3917` Eligible estimations for best-model selection: `3917`

Saved likelihoods are recomputed from the stored parameter paths before ranking. Large recursive shape states are evaluated by the BEGE saddlepoint density backend; `max(p_t, n_t)` is reported as a diagnostic, not as an exclusion rule.

Selection screen: finite corrected AIC/BIC/log-likelihood, successful optimizer status, finite positive shape paths, positive conditional variance paths, and documented parameter/stability/unconditional-variance constraints.

17.a. Global Best by AIC

- Mean type: `ARX(2,1)`
- Seed / draw: `91` / `3`
- AIC: `75.377840`
- BIC: `115.825497`
- LogLik: `-25.688920`
- Max shape: `395.826549`

17.b. Global Best by BIC

- Mean type: `ARX(2,1)`
- Seed / draw: `91` / `3`
- AIC: `75.377840`
- BIC: `115.825497`
- LogLik: `-25.688920`
- Max shape: `395.826549`

17.c. Eligible Best by Mean Type (AIC)

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	100	2	187.972717	214.937821	−85.986358	186.829178	0.072113
ARX(1,1)	72	4	186.399065	223.476083	−82.199533	204.131893	0.057609
ARX(2,1)	91	3	75.377840	115.825497	−25.688920	395.826549	0.035492
ARX(2,2)	34	2	291.884502	335.702796	−132.942251	146.783773	0.073678

17.d. Eligible Best by Mean Type (BIC)

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	100	2	187.972717	214.937821	−85.986358	186.829178	0.072113
ARX(1,1)	72	4	186.399065	223.476083	−82.199533	204.131893	0.057609
ARX(2,1)	91	3	75.377840	115.825497	−25.688920	395.826549	0.035492
ARX(2,2)	34	2	291.884502	335.702796	−132.942251	146.783773	0.073678

17.e. Parameter Estimates From Best AIC Fits

17.e.i. *constant:*

SE status: `computed`

Parameter	Estimate	Std. Error
p0	5.671685	0.968909
n0	0.005003	0.026126
rho_p	0.861405	0.002157
rho_n	0.117878	0.052076
phi_p	0.079471	0.000003
phi_n	0.881953	0.051794
sigma_p	0.072113	0.000005
sigma_n	0.287229	0.002115

17.e.ii. *ARX(1,1):*

SE status: `computed`

Parameter	Estimate	Std. Error
c	−0.134875	0.000316
rho_1	0.481958	0.000006
phi_1	1.067438	179.057216
p0	7.225714	975.382214
n0	6.429995	179.058119
rho_p	0.077997	179.065758
rho_n	0.955600	0.000380
phi_p	0.611996	0.000002
phi_n	0.012901	0.000239
sigma_p	0.313328	0.000321
sigma_n	0.057609	0.000283

17.e.iii. *ARX(2,1):*

SE status: `computed`

Parameter	Estimate	Std. Error
c	0.057319	6099.087934
rho_1	0.274752	6893.332675
rho_2	0.236062	1692.649286
phi_1	0.431935	8882.934120
p0	0.195254	0.057180
n0	5.063837	822793.344188
rho_p	0.998972	0.057781
rho_n	0.171948	10697.837383

Parameter	Estimate	Std. Error
phi_p	0.000010	0.000055
phi_n	0.635029	7535.413191
sigma_p	0.035492	0.057074
sigma_n	0.124464	0.000745

17.e.iv. *ARX(2,2):*

SE status: `computed`

Parameter	Estimate	Std. Error
c	0.057000	0.131283
rho_1	0.373407	0.739969
rho_2	0.188337	0.054558
phi_1	−0.330196	0.793610
phi_2	0.745016	0.077710
p0	5.273562	0.514440
n0	3.478176	2.063162
rho_p	0.727260	0.053210
rho_n	0.080459	0.360179
phi_p	0.070550	0.000002
phi_n	0.430884	0.054449
sigma_p	0.073678	0.000010
sigma_n	0.183330	0.000006

18. Inflation and Deflation Model

Here n_t, p_t is generated recursively with parameters $\left(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_n^-\right)$:

$$\begin{aligned} p_t &= p_0 + \rho_p\, p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2}\left(u_{t-1}^+\right)^2, \\ n_t &= n_0 + \rho_n\, n_{t-1} + \frac{\phi_n^-}{2\sigma_n^2}\left(u_{t-1}^-\right)^2 \end{aligned} \tag{67}$$

For the current random-search estimation runs, the likelihood is evaluated under the parameter bounds used in code, finite-recursion checks, and the documented ID-GARCH stability restrictions:

- $\rho_p + \frac{\phi_p^+}{2} < 1$ and $\rho_n + \frac{\phi_n^-}{2} < 1$.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \mathrm{Var}(\pi_t)$.

The hard shape cap $\max\{p_t, n_t\} < 200$ is **not imposed by default** in `ID_GARCH`. It can still be restored for sensitivity checks by passing `cap_pn=200`, but it is not part of the current best-model reporting screen.

For reported best-model selection, `collect_id_results.py` keeps the raw search rows, recomputes each likelihood with the stabilized BEGE density, and uses finite corrected criteria, optimizer convergence, and the documented parameter/stability/unconditional-variance constraints. The row-level selection outcome, stored likelihood, corrected likelihood, and maximum shape diagnostic are written to `results/selection_diagnostics.csv`.

19. Estimation Workflow

The Slurm workflow in `SimulationID.sh` submits 100 seed jobs by default. Each seed estimates all four mean processes on the fixed effective sample `1969Q2--2022Q4`:

- Constant
- ARX(1,1)
- ARX(2,1)
- ARX(2,2)

The job runner skips standard-error calculations and saves one CSV row after each model fit so partial jobs leave recoverable output. The collector merges the raw CSV files, picks the corrected best results, writes `results/best_model.md`, and computes standard errors for the best AIC fit within each eligible mean process in `results/best_aic_with_se.csv`.

20. Inflation/Deflation BEGE-GJR Best Model Summary

Generated: `2026-05-31T16:27:33` Total estimations: `702` Converged estimations: `73` Eligible estimations for best-model selection: `7`

Saved likelihoods are recomputed from the stored parameter paths before ranking. Large recursive shape states are evaluated by the BEGE saddlepoint density backend; `max(p_t, n_t)` is reported as a diagnostic, not as an exclusion rule.

Selection screen: finite corrected AIC/BIC/log-likelihood, successful optimizer status, finite positive shape paths, positive conditional variance paths, and documented parameter/stability/unconditional-variance constraints.

20.a. Global Best by AIC

- Mean type: `constant`
- Seed / draw: `54` / `2`
- AIC: `446.798303`
- BIC: `473.763407`
- LogLik: `-215.399151`
- Max shape: `62.877897`

20.b. Global Best by BIC

- Mean type: `constant`
- Seed / draw: `54` / `2`
- AIC: `446.798303`
- BIC: `473.763407`
- LogLik: `-215.399151`
- Max shape: `62.877897`

20.c. Eligible Best by Mean Type (AIC)

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	54	2	446.798303	473.763407	−215.399151	62.877897	0.213622
ARX(1,1)	92	2	510.492518	547.569537	−244.246259	19369.989267	0.009492
ARX(2,2)	55	1	553.766438	597.584732	−263.883219	43.635811	0.004287

20.d. Eligible Best by Mean Type (BIC)

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	54	2	446.798303	473.763407	−215.399151	62.877897	0.213622
ARX(1,1)	92	2	510.492518	547.569537	−244.246259	19369.989267	0.009492
ARX(2,2)	55	1	553.766438	597.584732	−263.883219	43.635811	0.004287

20.e. Parameter Estimates From Best AIC Fits

20.e.i. *constant:*

SE status: `computed`

Parameter	Estimate	Std. Error

Parameter	Estimate	Std. Error
p0	8.506759	1.583037
n0	0.005000	0.005161
rho_p	0.000010	0.259706
rho_n	0.000010	1.087155
phi_p_plus	1.050974	1.168389
phi_n_minus	0.000010	0.039698
sigma_p	0.213622	0.073689
sigma_n	0.463986	0.250562

20.e.ii. *ARX(1,1):*

SE status: `computed`

Parameter	Estimate	Std. Error
c	0.006195	0.083565
rho_1	0.226001	0.161346
phi_1	0.678940	0.183477
p0	6.714397	17.234445
n0	1.728725	0.426538
rho_p	0.313153	0.129167
rho_n	0.013862	0.234585
phi_p_plus	0.715508	0.098501
phi_n_minus	0.880822	1.096451
sigma_p	0.009492	0.001731
sigma_n	0.564762	0.237184

20.e.iii. *ARX(2,2):*

SE status: `computed`

Parameter	Estimate	Std. Error
c	0.055768	0.134788
rho_1	0.231099	0.126895
rho_2	0.333519	0.088086
phi_1	0.791746	0.384160
phi_2	−0.193110	0.307943
p0	1.555467	0.310893
n0	2.111224	0.506625
rho_p	0.000010	0.546156

Parameter	Estimate	Std. Error
rho_n	0.288606	0.131727
phi_p_plus	0.000010	0.108792
phi_n_minus	1.252742	0.292665
sigma_p	0.004287	0.004926
sigma_n	0.572866	0.144670

21. Full BEGE

BEGE GARCH has mean process and higher:

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p\, \omega_p - \sigma_n\, \omega_n, \end{aligned} \tag{68}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1). \tag{69}$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $\left(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^-\right)$:

$$\begin{aligned} p_t &= p_0 + \rho_p\, p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2}\left(u_{t-1}^+\right)^2 + \frac{\phi_p^-}{2\sigma_p^2}\left(u_{t-1}^-\right)^2, \\ n_t &= n_0 + \rho_n\, n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2}\left(u_{t-1}^+\right)^2 + \frac{\phi_n^-}{2\sigma_n^2}\left(u_{t-1}^-\right)^2 \end{aligned} \tag{70}$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$,
- $10^{-5} < \rho_p, \rho_n < 0.999$,
- $10^{-5} < \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^- < 0.999$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

According to the emails, I have set the constraints below.

- $\rho + \frac{\phi^+}{2} + \frac{\phi^-}{2} < 1$ for both BE and GE shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \mathrm{Var}(\pi_t)$ where $\mathrm{Var}(\pi_t) = 0.75$ is the unconditional variance of inflation.

The hard cap $\max\{p_t, n_t\} < 200$ is not imposed during raw multi-start optimization or reported best-model selection by default. The stabilized BEGE density is evaluated directly as long as the recursive shape series are finite; large shape states use the saddlepoint density backend. The collector writes the maximum shape path as a diagnostic in `results/selection_diagnostics.csv`.

22. Batch Estimation

`BEGE_GJR1.py` estimates the Full BEGE specification on the canonical effective sample from `DataSummary/README.md`, **1969Q2–2022Q4** with **215 observations**. The runner uses the precomputed lag columns when they are available, so lagged ARX regressors do not trim the sample again.

The script estimates the four mean-process specifications:

- Constant
- ARX(1,1)
- ARX(2,1)
- ARX(2,2)

Results are checkpointed to `output/raw/draw_###.csv` after each draw. This keeps completed mean-process rows when a seed job is interrupted, while the final write still contains all requested rows when the seed finishes.

`collect_full_results.py` merges the raw seed files and writes:

- `results/all_estimations.csv`, without standard-error columns or optimizer messages.
- `results/selection_diagnostics.csv`, with stored and corrected likelihood criteria plus selection diagnostics.
- `results/best_aic_with_se.csv`, with standard errors for the corrected best AIC fit in each eligible mean process.
- `results/best_model.md`, with the corrected best AIC/BIC models and reported parameter standard errors.

When `START_ID` and `END_ID` are set, the collector only merges `draw_###.csv` files in that seed range. This prevents older seed files from entering a new smaller resubmission.

`SimulationGJR.sh` defaults to 400 seed jobs, 10 draws per mean process, 10 optimizer starts per draw, and 300 SLSQP iterations per start. This gives 40,000 optimizer starts for each mean process. With all four mean processes, that is 160,000 optimizer starts total.

23. Full BEGE Best Model Summary

Generated: `2026-05-31T16:29:30` Total estimations: `4004` Converged estimations: `4004` Eligible estimations for best-model selection: `4004`

Saved likelihoods are recomputed from the stored parameter paths before ranking. Large recursive shape states are evaluated by the BEGE saddlepoint density backend; `max(p_t, n_t)` is reported as a diagnostic, not as an exclusion rule.

Selection screen: finite corrected AIC/BIC/log-likelihood, successful optimizer status, finite positive shape paths, positive conditional variance paths, and documented parameter/stability/unconditional-variance constraints.

23.a. Global Best by AIC

- Mean type: `ARX(2,2)`
- Seed / draw: `7` / `8`
- AIC: `-6.730114`
- BIC: `43.829457`
- LogLik: `18.365057`
- Max shape: `745.568795`

23.b. Global Best by BIC

- Mean type: `ARX(2,2)`
- Seed / draw: `7` / `8`
- AIC: `-6.730114`
- BIC: `43.829457`
- LogLik: `18.365057`
- Max shape: `745.568795`

23.c. Eligible Best by Mean Type (AIC)

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	90	10	107.459755	141.166135	−43.729878	360.905485	0.073864
ARX(1,1)	3	10	275.192039	319.010334	−124.596020	226.890218	0.078104
ARX(2,1)	97	1	196.337067	243.525999	−84.168534	187.521810	0.096691
ARX(2,2)	7	8	−6.730114	43.829457	18.365057	745.568795	0.053275

23.d. Eligible Best by Mean Type (BIC)

Mean Type	Seed	Draw	AIC	BIC	LogLik	Max Shape	Min Sigma
constant	90	10	107.459755	141.166135	−43.729878	360.905485	0.073864
ARX(1,1)	3	10	275.192039	319.010334	−124.596020	226.890218	0.078104
ARX(2,1)	97	1	196.337067	243.525999	−84.168534	187.521810	0.096691
ARX(2,2)	7	8	−6.730114	43.829457	18.365057	745.568795	0.053275

23.e. Parameter Estimates From Best AIC Fits

23.e.i. *constant:*

SE status: `computed`

Parameter	Estimate	Std. Error
p0	1.020568	642.603477
n0	2.629179	1930.516048
rho_p	0.000140	467.719148
rho_n	0.284566	22.302143
phi_p_plus	0.060659	960.348389
phi_p_minus	0.999000	12.869956
phi_n_plus	0.732789	227.103516
phi_n_minus	0.118977	0.000416
sigma_p	0.299363	0.000409
sigma_n	0.073864	0.000412

23.e.ii. *ARX(1,1):*

SE status: `computed`

Parameter	Estimate	Std. Error
c	−0.108817	3.723452
rho_1	0.401823	8.700547
phi_1	0.647492	16.436002
p0	6.027018	7.410728
n0	4.401859	5.342318
rho_p	0.222627	0.538293
rho_n	0.230459	0.157217
phi_p_plus	0.512699	0.211133
phi_p_minus	0.112869	0.000041
phi_n_plus	0.127898	3.001225
phi_n_minus	0.769526	2.173598
sigma_p	0.078104	0.000000
sigma_n	0.255866	0.386044

23.e.iii. *ARX(2,1):*

SE status: `computed`

Parameter	Estimate	Std. Error
c	0.112878	0.000491
rho_1	0.219899	0.472785
rho_2	0.123548	0.000479
phi_1	0.569001	0.473256

Parameter	Estimate	Std. Error
p0	1.214025	3309.782984
n0	3.434311	1434.096076
rho_p	0.252028	1434.100350
rho_n	0.552227	0.000000
phi_p_plus	0.267015	9411.917053
phi_p_minus	0.822841	0.000475
phi_n_plus	0.272032	0.000000
phi_n_minus	0.179901	0.000497
sigma_p	0.201014	0.000473
sigma_n	0.096691	0.000528

23.e.iv. *ARX(2,2):*

SE status: `computed`

Parameter	Estimate	Std. Error
c	−0.017255	60.223484
rho_1	0.001986	15.600399
rho_2	0.059976	32.980582
phi_1	0.213953	116.842533
phi_2	1.913764	105.692778
p0	5.608108	3927.875724
n0	6.078626	2153.442460
rho_p	0.228366	3.189194
rho_n	0.636047	40.359291
phi_p_plus	0.324595	2413.806029
phi_p_minus	0.183621	5.177028
phi_n_plus	0.063376	79.566406
phi_n_minus	0.505966	0.321695
sigma_p	0.053275	0.001003
sigma_n	0.296416	3.459597